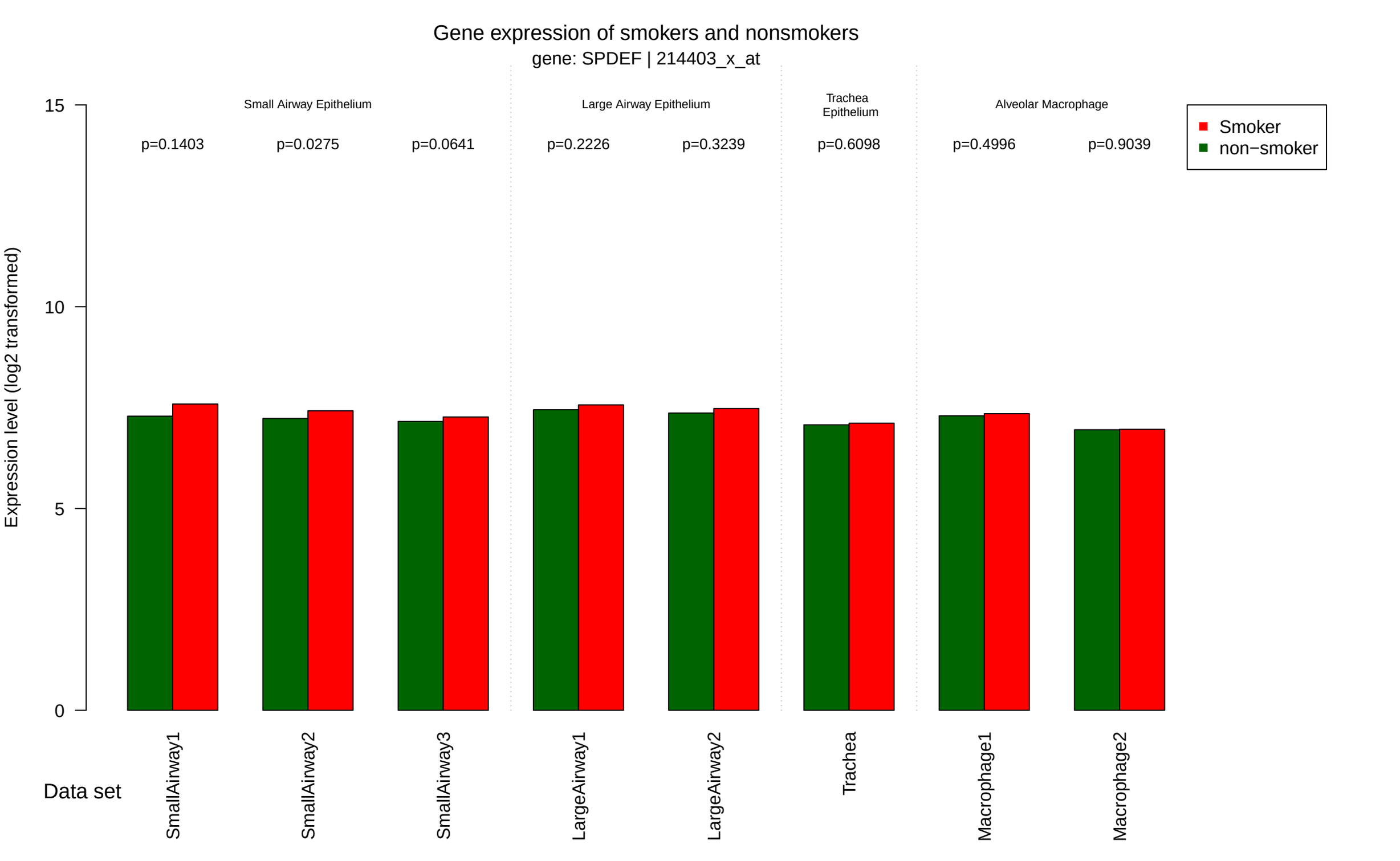
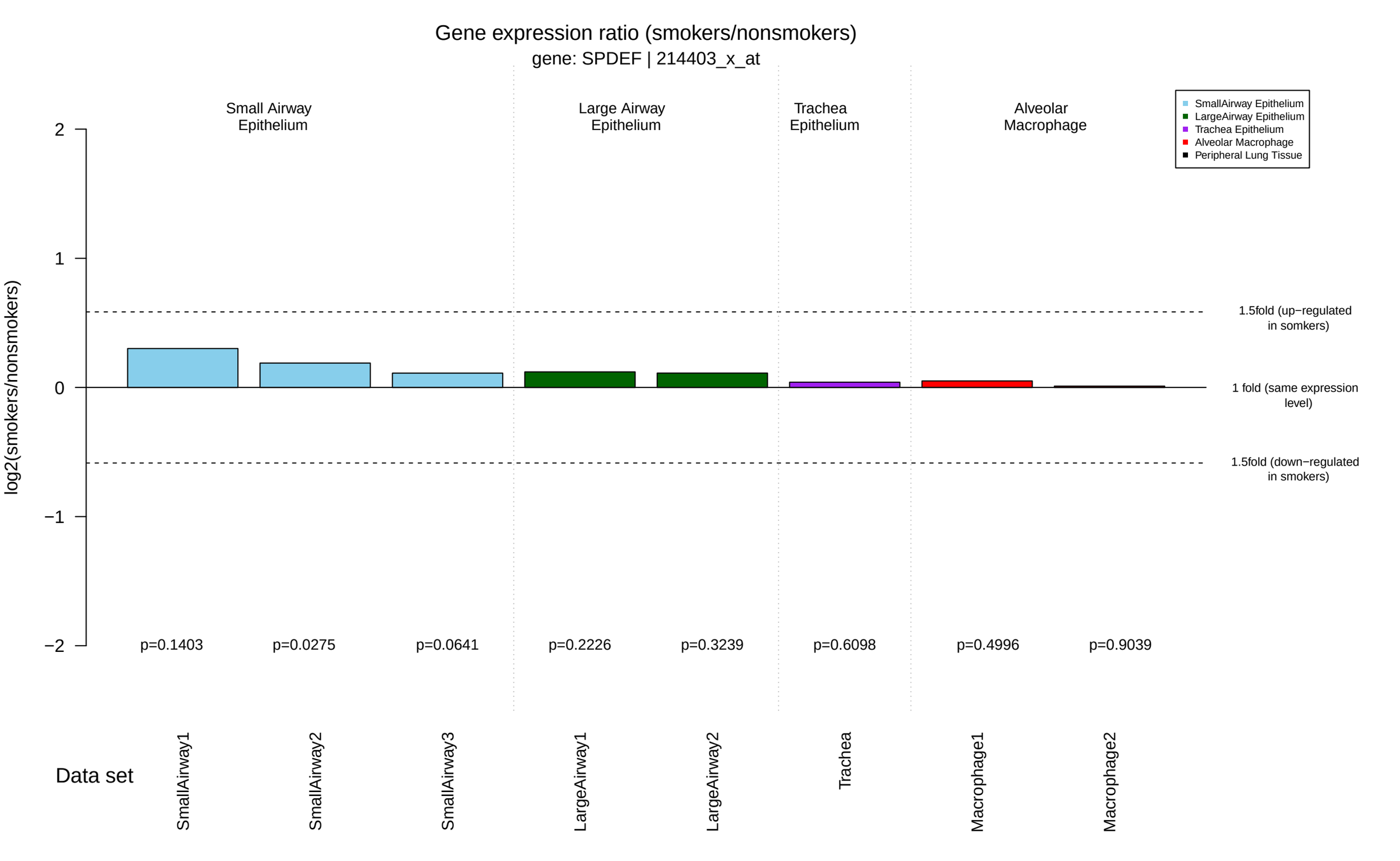
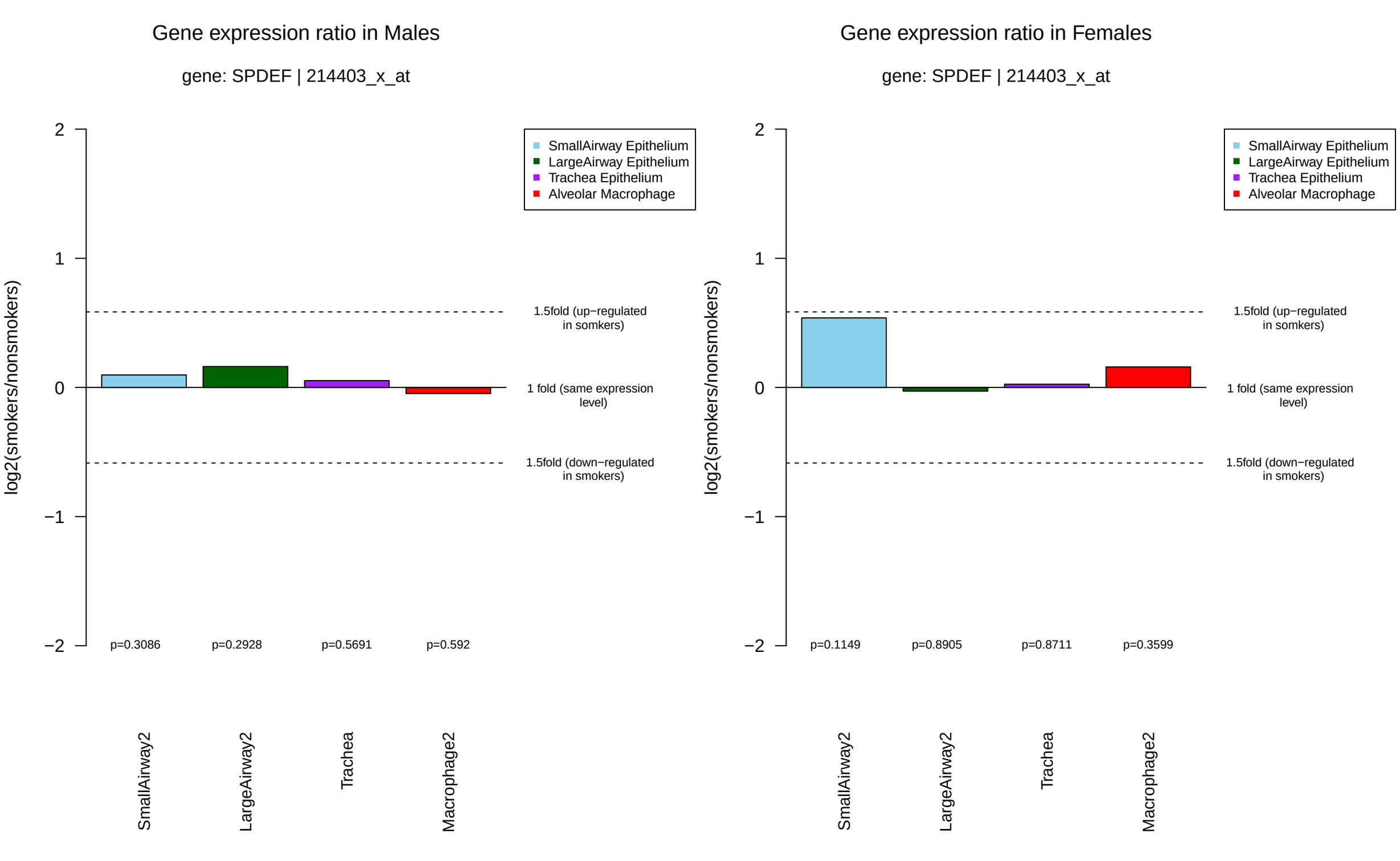
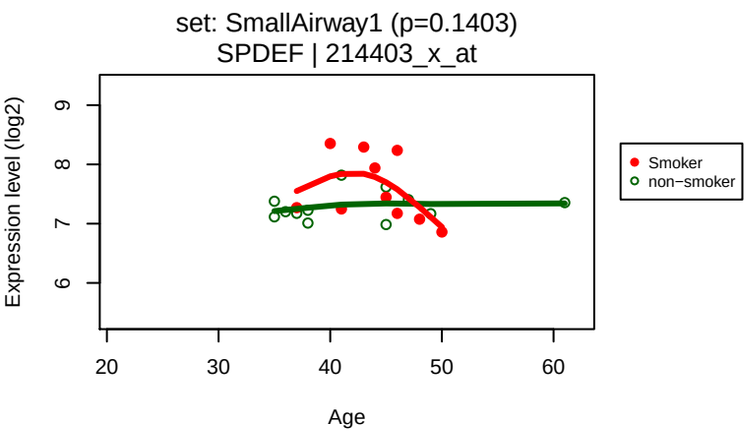
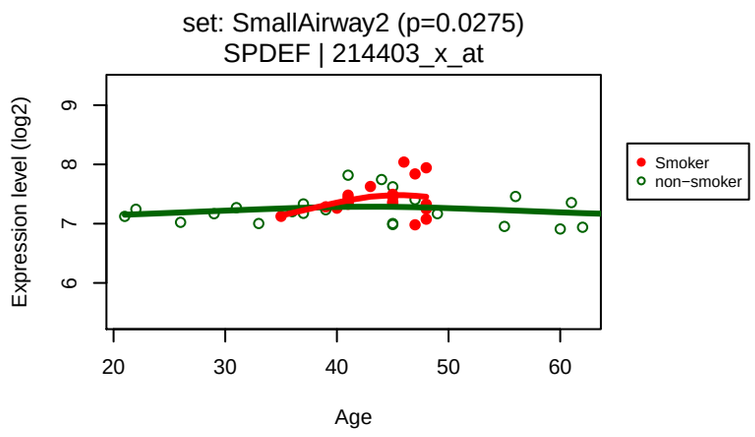
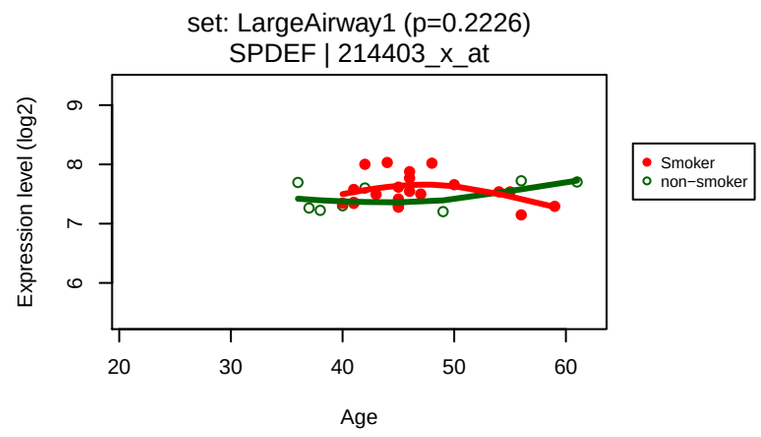

Gene expression of smokers and nonsmokers
gene: SPDEF | 214403_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1403
p=0.0275
p=0.0641
p=0.2226
p=0.3239
p=0.6098
p=0.4996
p=0.9039
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

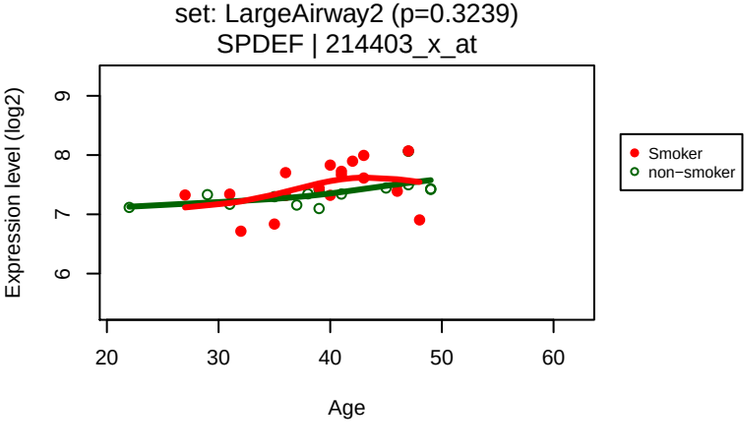

Gene expression ratio (smokers/nonsmokers)
gene: SPDEF | 214403_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1403
p=0.0275
p=0.0641
p=0.2226
p=0.3239
p=0.6098
p=0.4996
p=0.9039
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

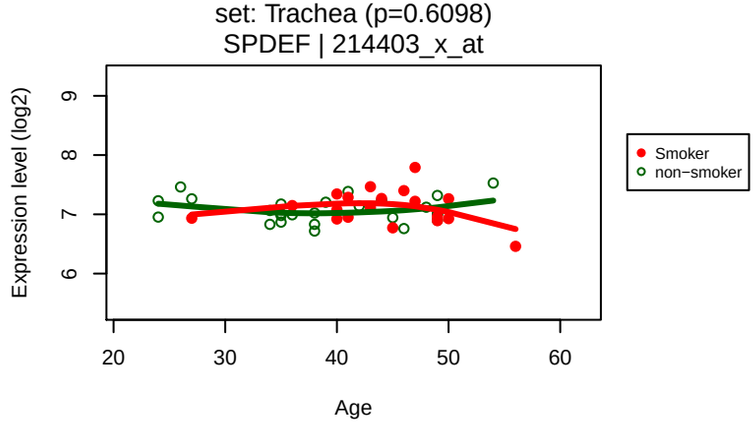

Gene expression ratio in Males
gene: SPDEF | 214403_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3086
p=0.2928
p=0.5691
p=0.592
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPDEF | 214403_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1149
p=0.8905
p=0.8711
p=0.3599
SmallAirway2
LargeAirway2
Trachea
Macrophage2

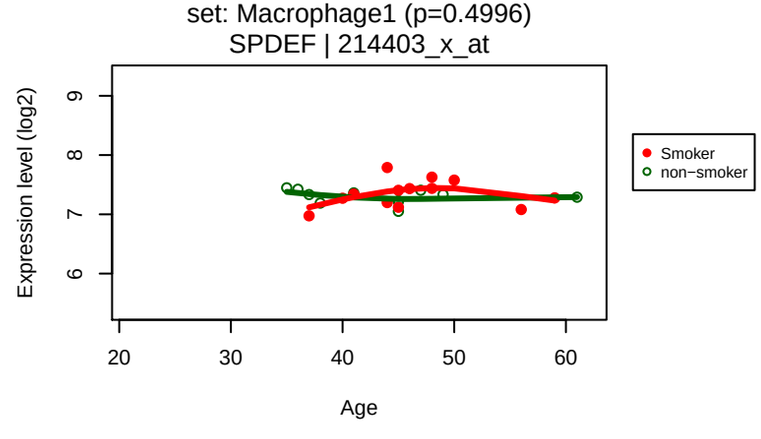

set: SmallAirway1 (p=0.1403)
SPDEF | 214403_x_at
Expression level (log2)
Age
Smoker
non−smoker

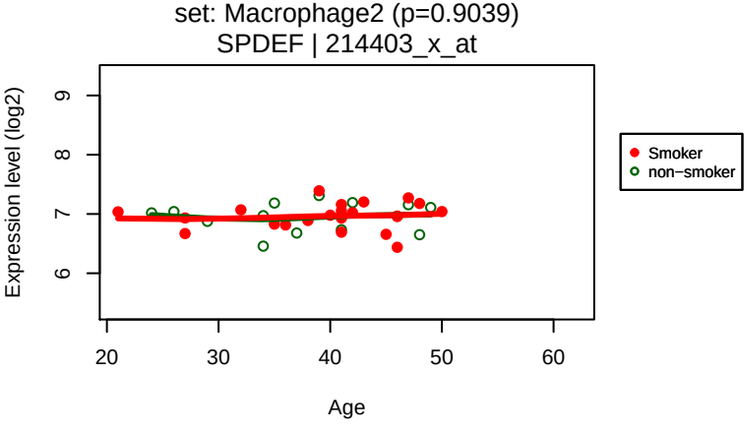

set: SmallAirway2 (p=0.0275)
SPDEF | 214403_x_at
Expression level (log2)
Age
Smoker
non−smoker

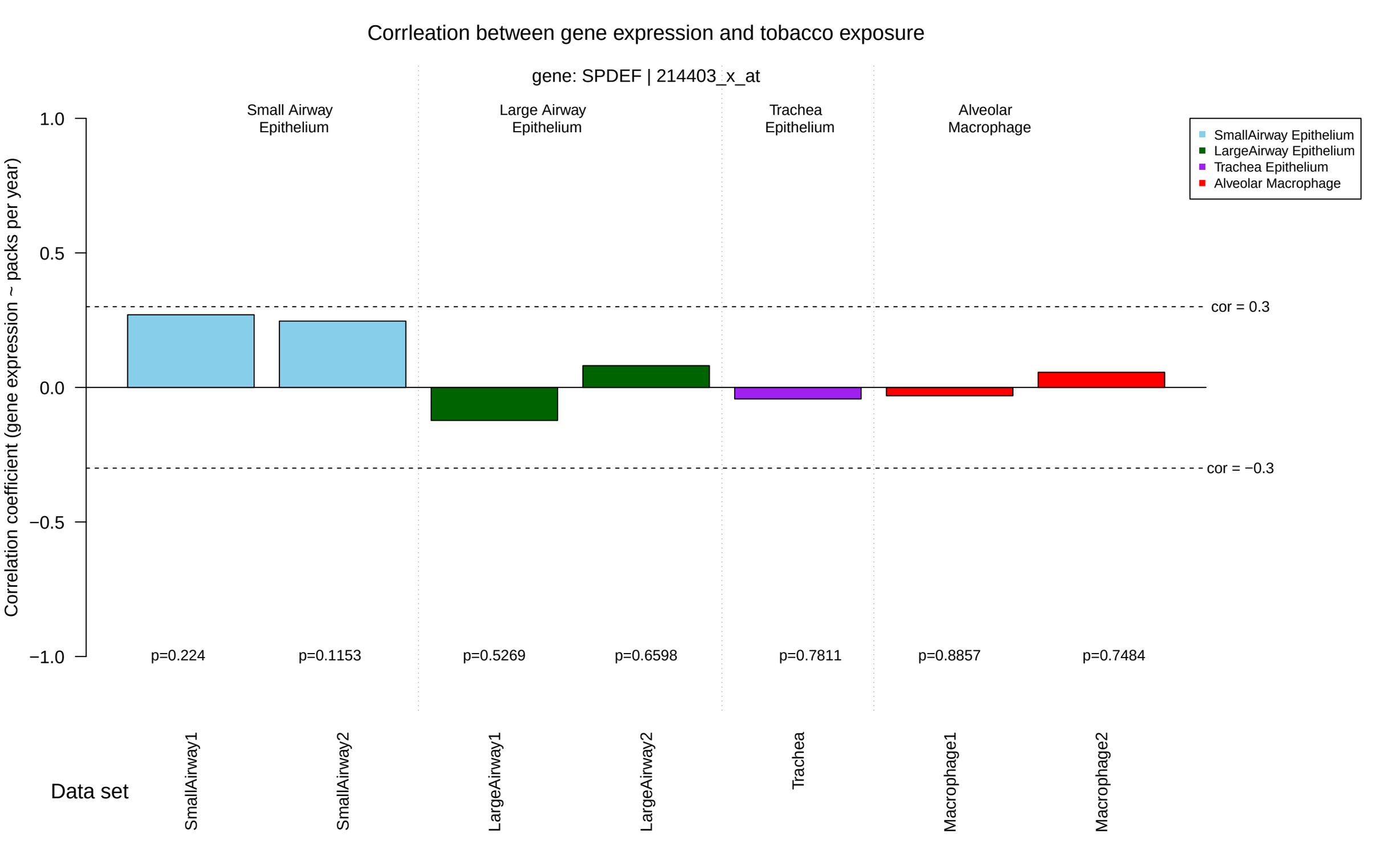

set: LargeAirway1 (p=0.2226)
SPDEF | 214403_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3239)
SPDEF | 214403_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6098)
SPDEF | 214403_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4996)
SPDEF | 214403_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9039)
SPDEF | 214403_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPDEF | 214403_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.224
p=0.1153
p=0.5269
p=0.6598
p=0.7811
p=0.8857
p=0.7484
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2